

Mount Kenya University



Unlocking Infinite Possibilities

STUDENT NAME: COLLINS MATAGI

ADMISSION NUMBER: BSCCS/2024/50482

COURSE/CLASS: COMPUTER SCIENCE 2024 SEP CLASS

UNIT CODE: BIT3208

UNIT: ADVANCED WEB DESIGN AND DEVELOPMENT

LECTURE NAME: MICHAEL NYORO

PROJECT TITLE: Apex Library Management System.

TECHNOLOGIES USED:

- **Backend Engine:** PHP for session validation, and circulation calculations.
- **Database Engine:** MySQL.

GITHUB LINK: <https://github.com/t4g1z>

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WEEK ONE: Xampp Local Environment Setup

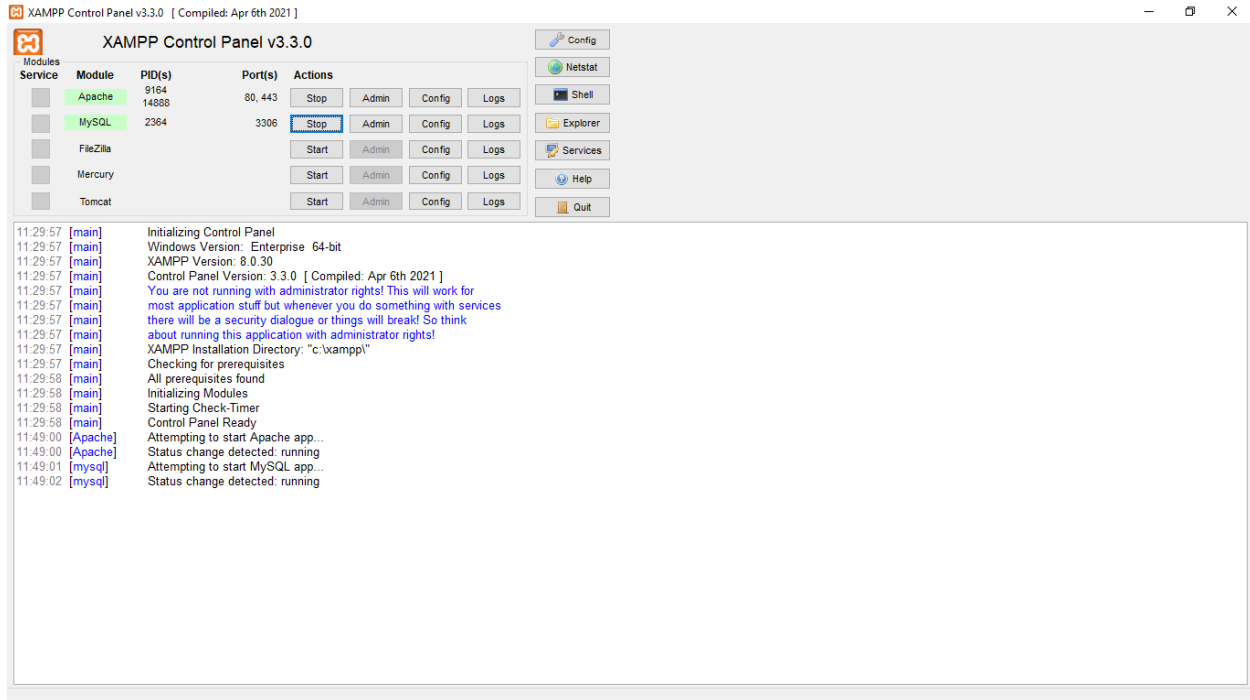


Fig 1.1 Apache and MySQL Running

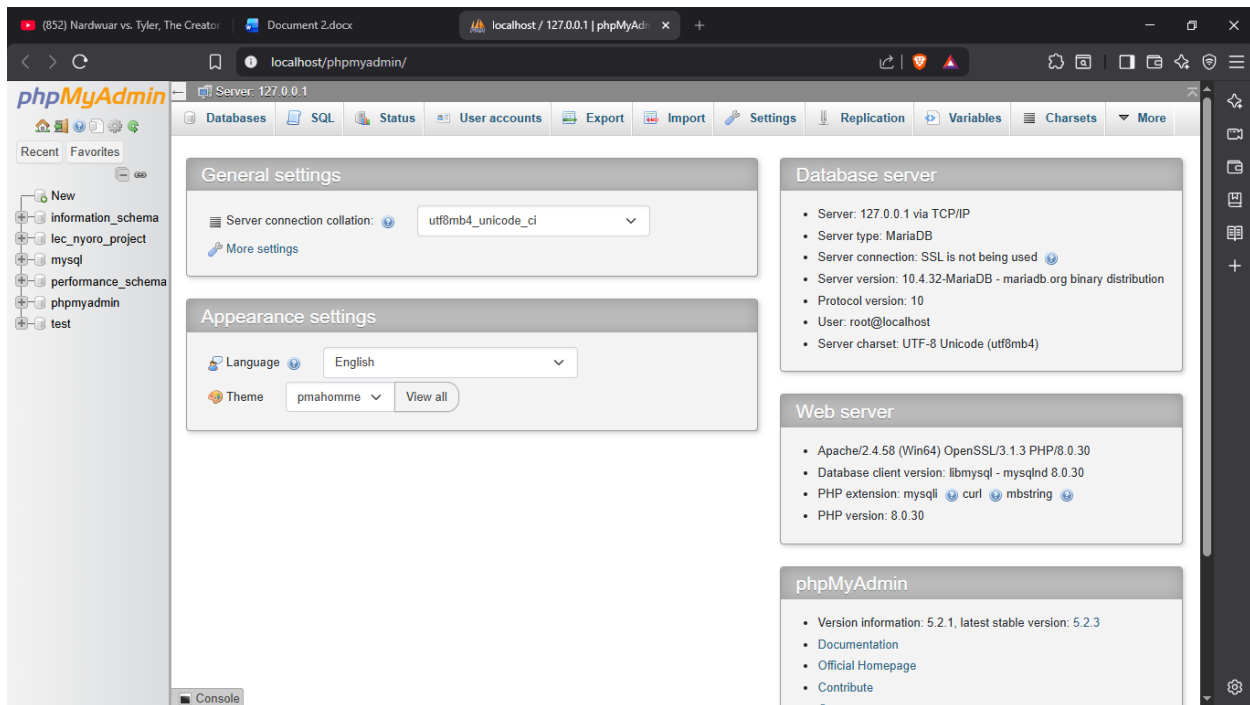


Fig 1.2 Local host test page

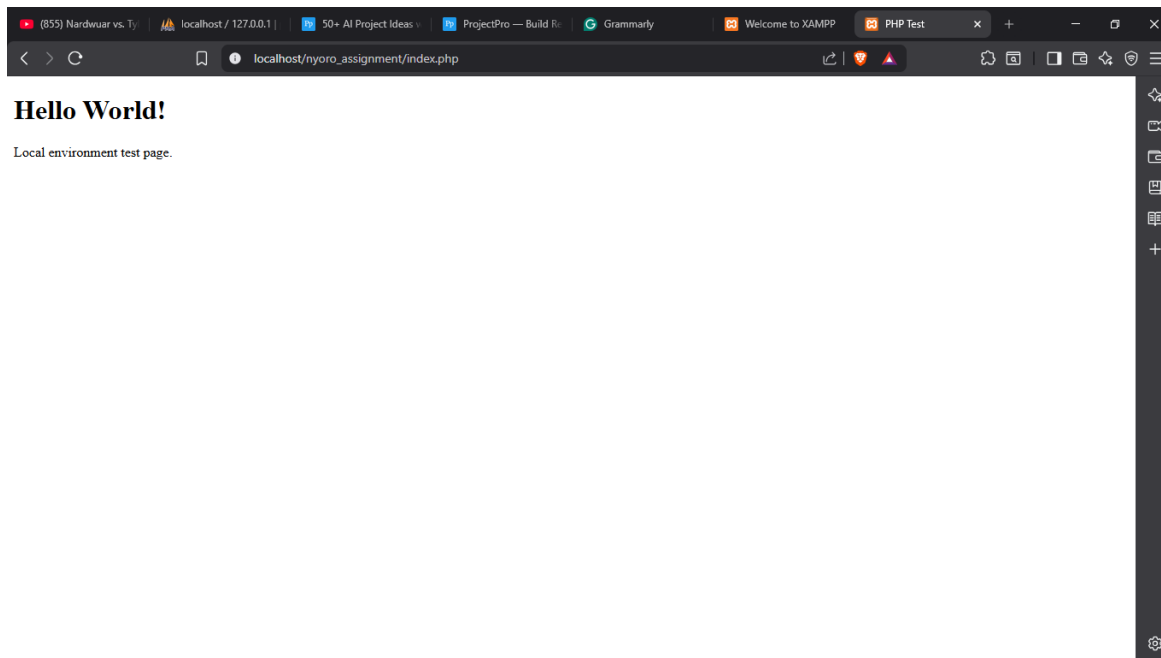


Fig 1.3 Mock page displaying HELLO WORLD

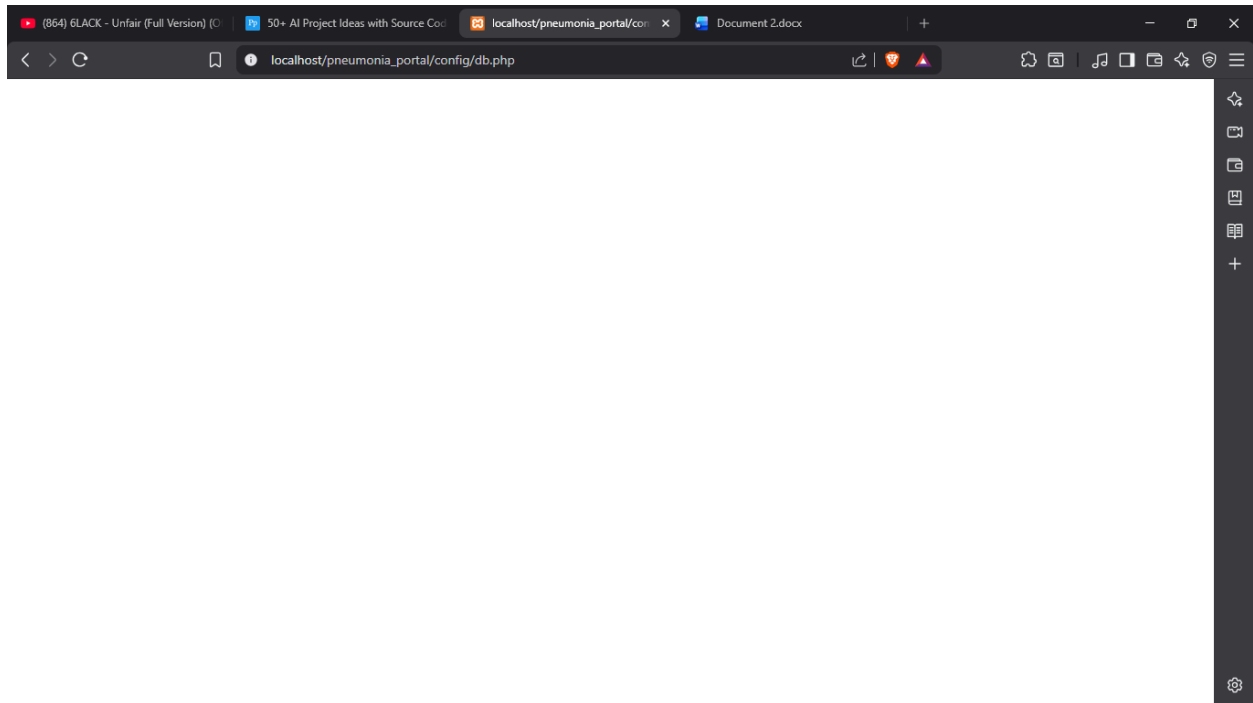


Fig 1.4 Shows a blank page to confirm that the connection to MySQL has no errors and is connected successfully.

What i learnt

Mastered local server architecture deployment by configuring XAMPP (Apache and MySQL). Learned the difference between client-side and server-side execution and organized a modular, professional project directory structure for proper software development workflow.

WEEK 2: UI/UX AND DESIGN FOUNDATIONS

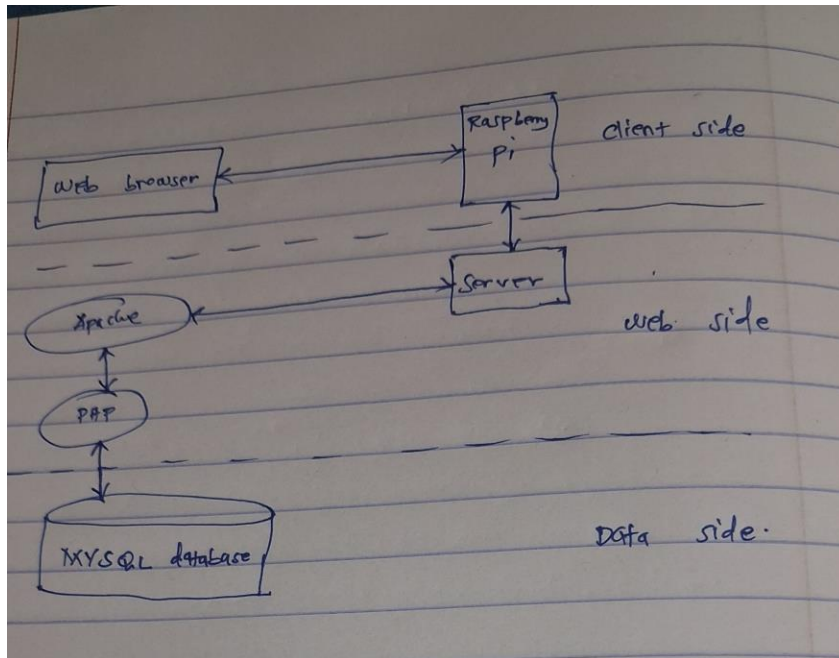


Fig 2.1 Client-Server Data Distribution Architecture Model.

The wireframe shows a login/registration interface for the 'Apex Library System' (SYSTEM GATEWAY ROUTER). It includes the following elements:

- Header:** Apex Library System logo and title.
- Buttons:** 'Account Log In' and 'Register Card'.
- Form Fields:**
 - 'Account Context Role' dropdown menu (selected: Student Borrower Portal).
 - 'Library Card ID Number' text input (placeholder: e.g. STU-2024-1547).
 - 'Security Profile Password' text input (masked with dots).
- Action Button:** 'Authenticate Secure Access'.
- Side Bar:** A vertical toolbar on the right with icons for zooming, panning, and other design tools.

Fig 2.2 Figma wireframe to show the login/registration form.

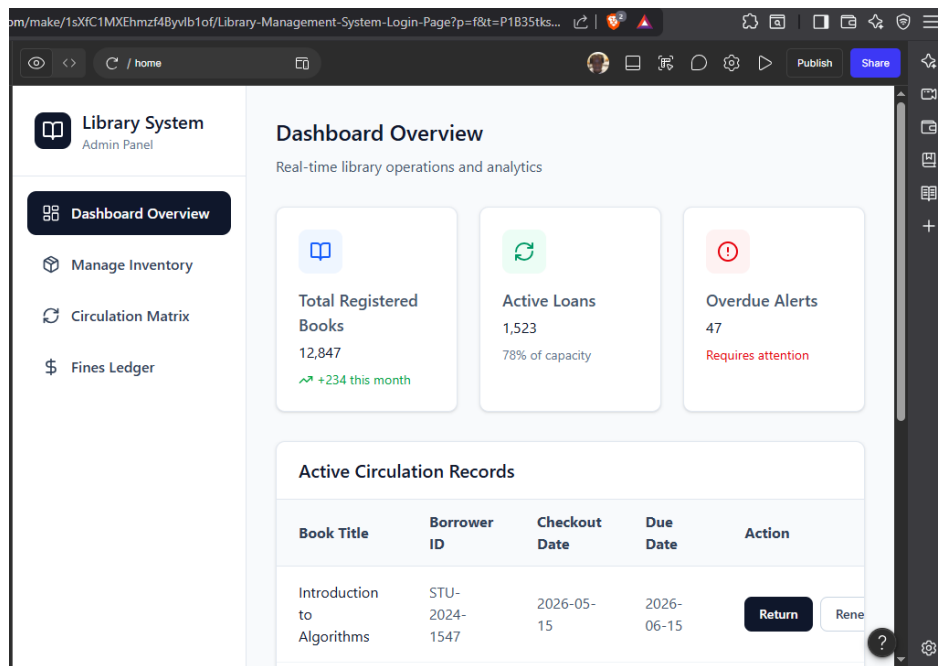


Fig 2.3 Figma wireframe to show dashboard.

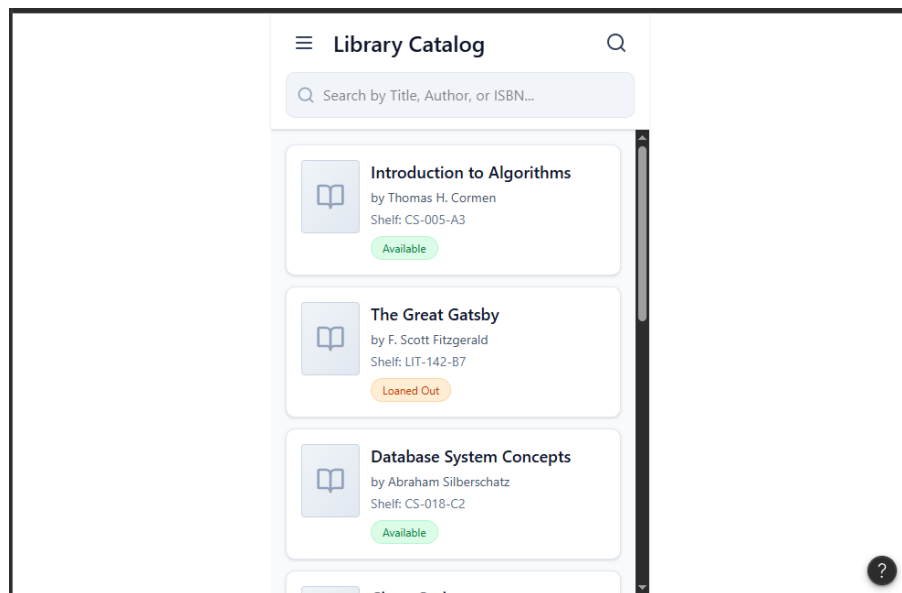


Fig 2.4

Fig 2.2, 2.3 and 2.4 show Figma wireframe designed for the intended system.

What i learnt

Explored design thinking and user experience paradigms by creating low-fidelity UI/UX structural wireframes in Figma. Developed responsive interfaces for desktop and mobile form screens, utilizing standard layout placeholder notations to map out the application's visual flow.

WEEK 3: FRONTEND SCRIPTING AND BACKEND PROGRAMMING FOUNDATIONS

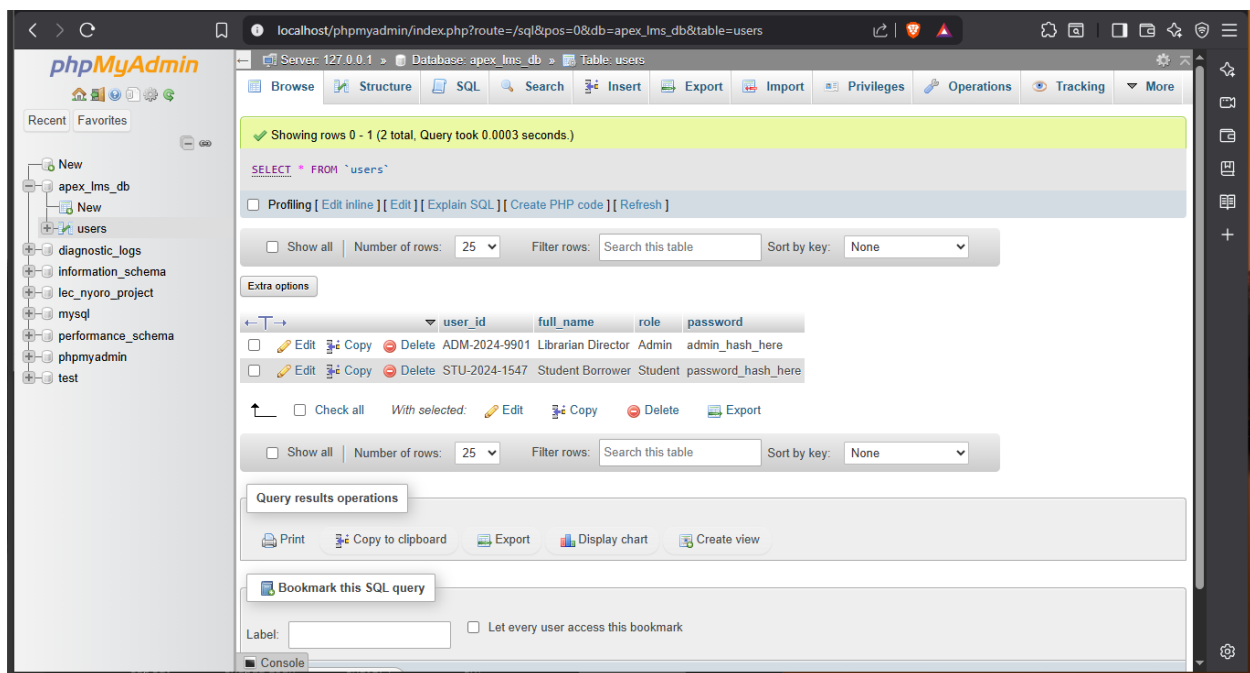


Fig 3.1 Shows my database for apex library management system.

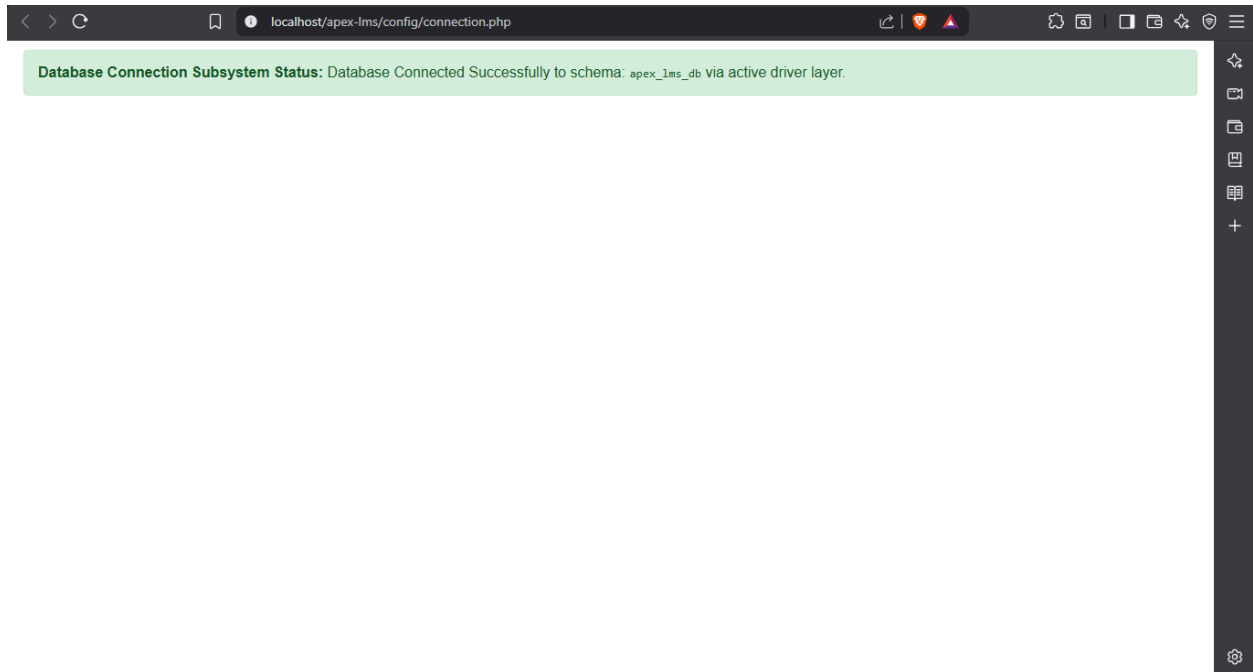


Fig 3.2 Shows that the database is connected successfully.

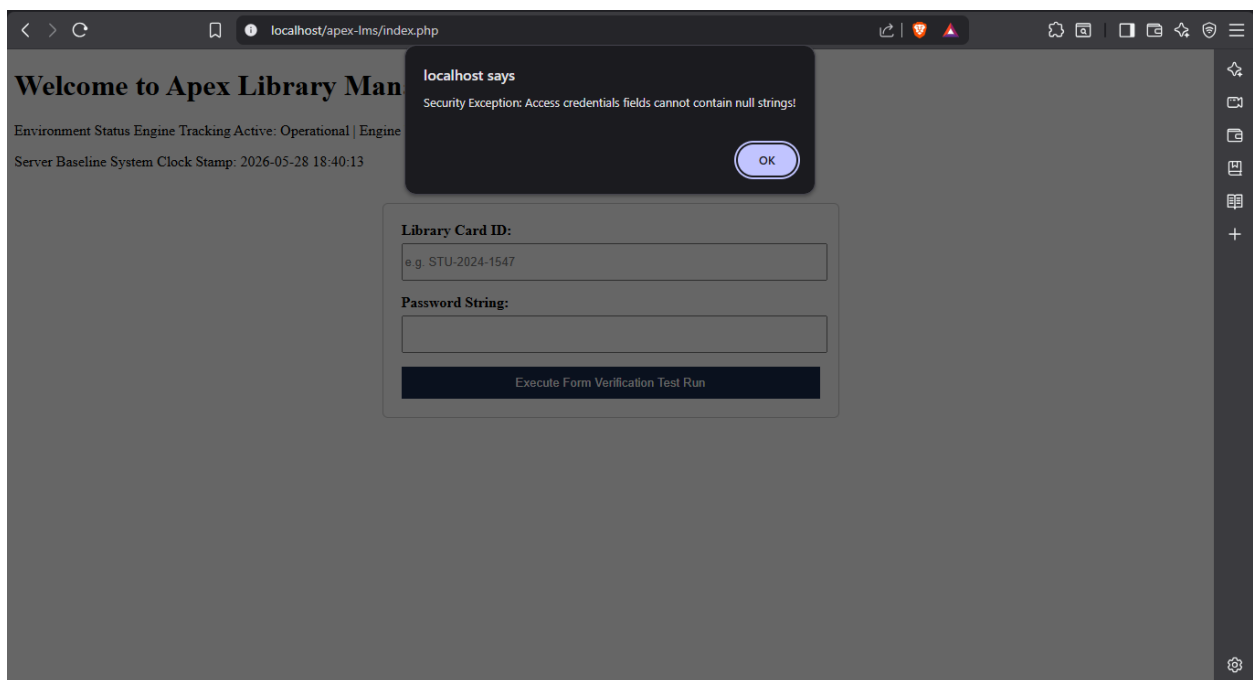


Fig 3.3 Shows error when the login page has nothing inserted in the fields.

What i learnt

Introduced client-side scripting and backend foundations. Built a form validation system using JavaScript DOM event manipulation, implemented a real-time password strength analyzer, and established a secure, object-oriented database connection bridge using modern PHP Data Objects (PDO) strings.

GITHUB VERSION CONTROL 1: <https://github.com/t4g1z/apex-lib>

WEEK 4: DYNAMIC BACKEND FORM PROCESSING & SYSTEM AUTHENTICATION

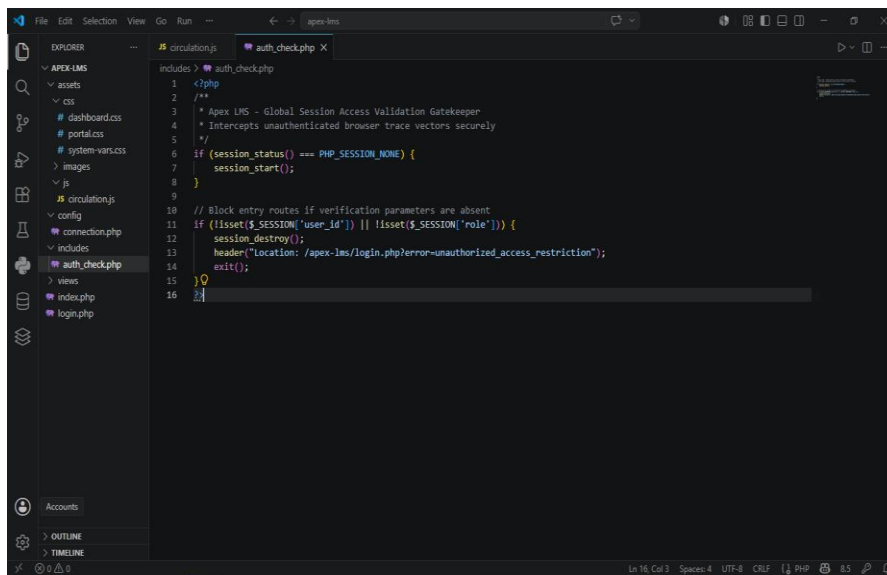


fig 4.1 Security Gatekeeper Middleware.

```
1 <?php
2 /**
3  * Apex LMS - Secure Gateway Portal Core Routing Controller
4  * Ref Layout Context: Figure 2.2 Structural Blueprint
5  */
6 // 1. Initialize safe stateful session data engine tracking channels
7 if (session_status() === PHP_SESSION_NONE) {
8     session_start();
9 }
10
11 // 2. Import central structural database connection instance handles
12 require_once __DIR__ . '/config/connection.php';
13
14 $authErrorMessage = "";
15
16 // 3. Intercept inbound POST data payloads safely
17 if ($_SERVER["REQUEST_METHOD"] == "POST") {
18     // Sanitize inbound string sequences against basic script injection attempts
19     $inputCardId = htmlspecialchars(trim($_POST['libraryCardId']));
20     $inputPassword = trim($_POST['userPassword']);
21     $selectedRole = htmlspecialchars(trim($_POST['userRole']) ?? 'Student');
22
23     if (empty($inputCardId) || empty($inputPassword)) {
24         $authErrorMessage = "Security Breach Reject: Form fields cannot be blank!";
25     } else {
26         try {
27             // Compile high-security parameterized database transaction queries
28             $queryText = "SELECT * FROM users WHERE user_id = :user_id AND role = :role LIMIT 1";
29             $stmt = $pdo->prepare($queryText);
30             $stmt->execute([
31                 ':user_id' => $inputCardId,
32                 ':role' => $selectedRole
33             ]);
34         } catch (PDOException $e) {
35             // Handle database connection errors
36         }
37     }
38 }
```

Fig 4.2 Login page form code.

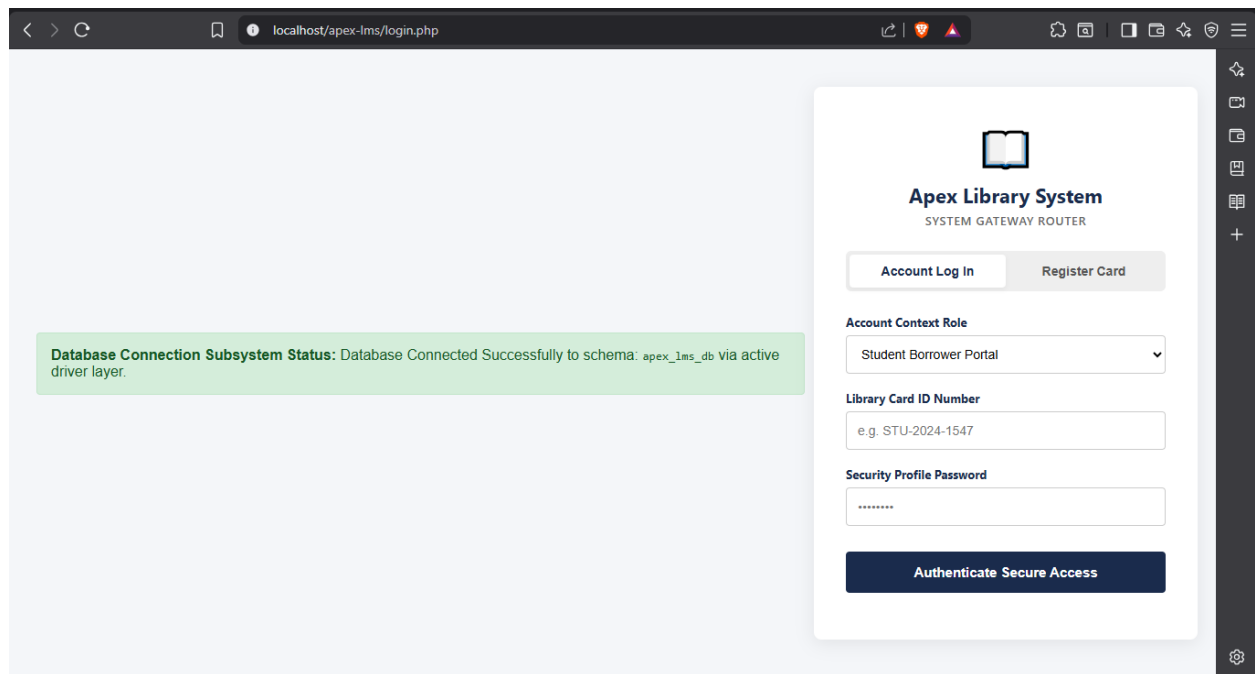


Fig 4.3 Login page active.

What i learned

Learned the full HTTP request-response cycle by integrating HTML forms with backend processing. Implemented secure data handling via the POST method, introduced user authentication pipelines using session tracking (\$_SESSION), and created a defensive middleware gatekeeper file.

WEEK 5: DATABASE COMPLETION AND DYNAMIC CRUD SYSTEM LAYER INTEGRATION

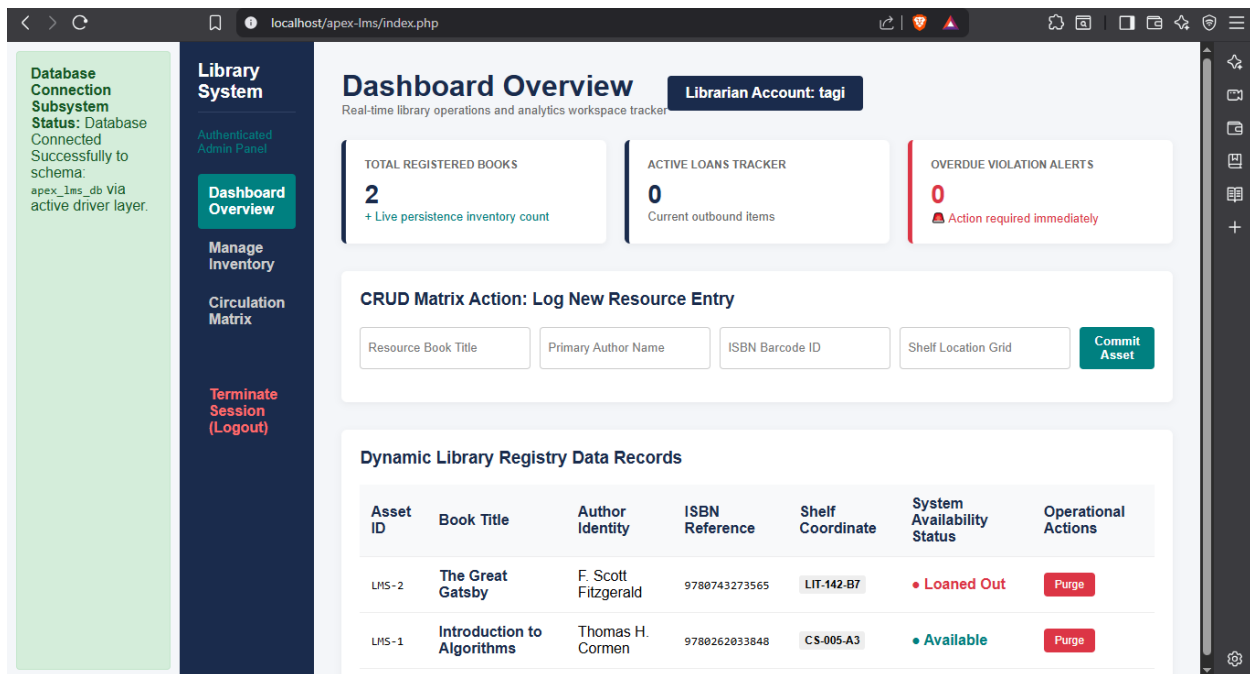


Fig 5.1 Shows the final dashboard page after login

What i learned

Implemented full backend data-driven integration. Constructed relational MySQL schemas with foreign key constraints, mastered full CRUD mechanics (Create, Read, Update, Delete) to dynamically add, display, and purge database assets, and effectively debugged system runtime errors like SQLSTATE base table exceptions.

GITHUB VERSION CONTROL 2: <https://github.com/t4g1z/apex-lib/tree/main/apex-lms>